



Lab 04

Sequentially Switched Circuits

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4.1 Objective

The objective of this lab is to examine and predict the response of sequentially switched RC and RL circuits.

Pre-Lab: Task 1

Q1. Complete the Table 4.1.

Q2. Explain Natural Response, Forced Response and Complete Response of an RL and RC Circuits. Give equations for all.

Q3. For Task 2 find the value of Tau and complete the calculation¹ section of Table 4.2 write maximum value of the capacitor charging voltage. Show your calculations.

Note: Do your Pre-Lab on a separate A4 sized sheet.
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4.2 Introduction

In Lab 1, the behavior of an RC and RL circuit at time $t = 0$ has been observed. In this Lab we are going to predict and analyze the behavior of RC and RL circuits when switching takes place at other times. In later Section 4.4 of this Lab, you will be introduced to the analysis of series and parallel RL and RC circuits.

4.3 Sequentially Switched RL and RC Circuits

In predicting the response of simple RL and RC circuits subjected to pulses and series of pulses sometimes referred to as sequentially switched circuits, the key is the relative size of the circuit time constant to the various times that define the pulse sequence. The underlying principle behind the analysis will be whether the energy storage element has time to fully charge before the pulse ends, and whether it has time to fully discharge before the next pulse begins.

Consider the circuit shown in Figure 4.1a, which is connected to a pulsed voltage source described by seven separate parameters defined in Figure 4.1b, which you are already familiar with. The waveform is bounded by two values, V1 and V2.

¹ Revise the capacitor charging and discharging topic already covered in Electric Circuits I & II courses, when using formula, assume that the capacitor charges completely i.e., 100%.

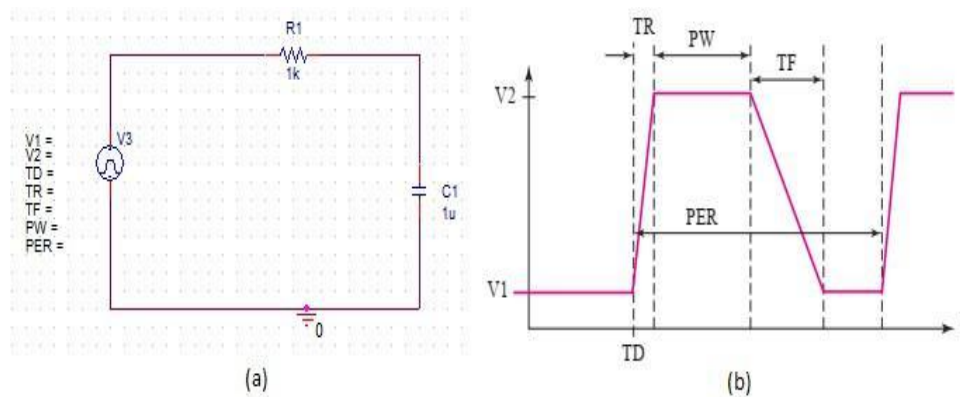


Figure 4.1: (a) Schematic of a simple RC circuit connected to pulse voltage waveform. (b) Diagram of the SPICE VPULSE parameter definitions

For the purposes of this discussion, we set a zero-time delay, $V1 = 0$, and $V2 = 9$ V. The circuit time constant is $\tau = RC = 1$ ms, so we set the rise and fall times to be 1 ns. Although SPICE will not allow a voltage to change in zero time since it solves the differential equations using discrete time intervals, compared to our circuit time constant 1 ns is a reasonable approximation to “instantaneous.” We will consider four basic cases, summarized in Table 4.1. In the first two cases, the input-pulse width (PW) is much longer than the circuit time constant τ , so we expect the transients resulting from the beginning of the pulse to die out before the pulse is over. In the latter two cases, the opposite is true: the pulse width is so short that the capacitor does not have time to fully charge before the pulse ends. A similar issue arises when we consider the response of the circuit when the time between pulses (PER - PW) is either short (Case II) or long (Case III) compared to the circuit time constant.

Table 4.1 - Four separate cases of pulse width and period relative to the circuit time constant of 1 ms

Case	Pulse Width (PW)	Period (PER)	(PER-PW) (2 Points)
I	10ms ($\tau \ll PW$)	20ms ($\tau \ll PER - PW$)	10ms
II	10ms ($\tau \ll PW$)	10.1ms ($\tau \gg PER - PW$)	0.1ms
III	0.1ms ($\tau \gg PW$)	10.1ms ($\tau \ll PER - PW$)	10ms
IV	0.1ms ($\tau \gg PW$)	0.2ms ($\tau \gg PER - PW$)	0.1ms

Task 2: To examine the response of RC Circuit

$$\tau_{\text{(Calculated)}} = 1\text{ms}$$

Pre-Lab (2 Points)

Case I: When the capacitor has enough time to charge or discharge

- Figure 4.1a shows the schematic for a simple RC circuit. Simulate the circuit using PSpice as per the case I in Table 4.1.
- Specify the “Run to Time” as 100ms to run the simulation for five full periods.



3. Since the capacitor has time to fully charge, the forced response will correspond to the 9V DC driving voltage. Write the **equation** for complete response of Case 1. The space for equation is given below

$$V_{\text{complete response}} = V_0 \exp^{-\frac{t}{\tau}} + V_s(1 - \exp^{-\frac{t}{\tau}})$$

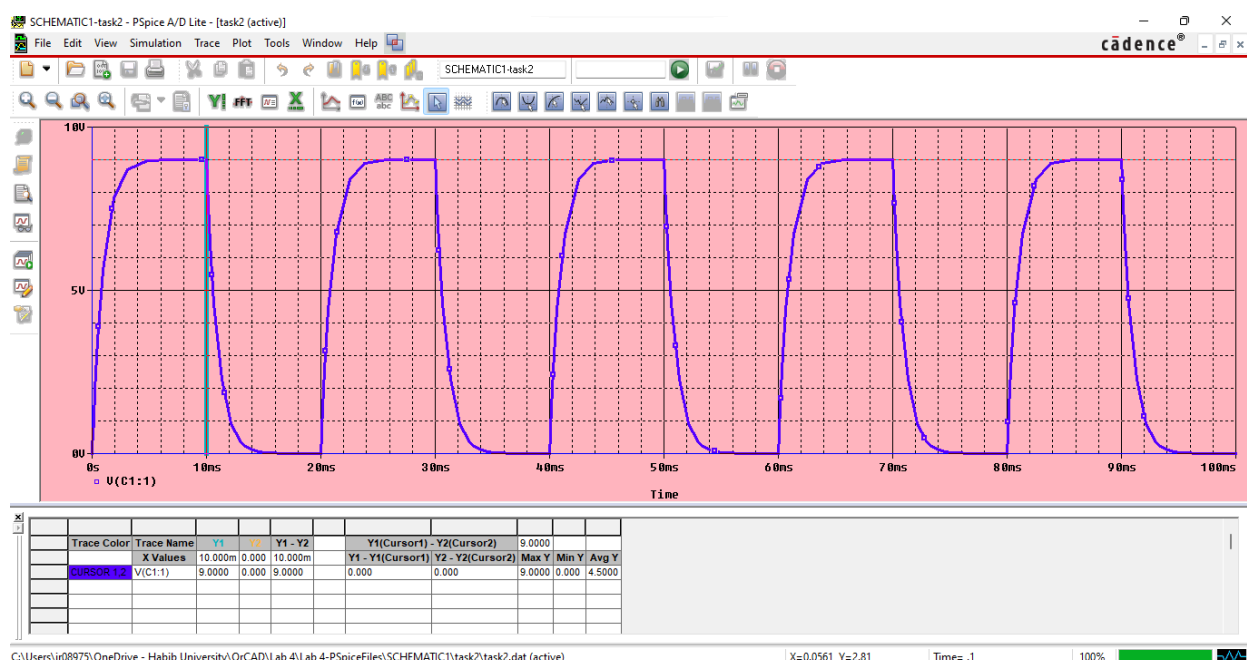
4. Attach the snippet of simulated circuit and plot of capacitor voltage. Plots should be visible and with white background.
5. Comment on the results with proper reasoning.

Space for Analysis:

Space for Analysis:

$$V_C(t) = 9 \times (1 - e^{-\frac{t}{\tau}}) \quad (V_0 = 8.9996 \text{ V})$$
$$= 9 \times (1 - e^{-10}) = 8.9996 \text{ V} \Rightarrow \text{charging}$$

comments: the PW is much longer than the τ . This allows the capacitor to fully charge to almost the driving voltage (9V). The slight difference between measured and calculated values is due to minor simulator tolerance, but otherwise proves our calculations.



Case II: When the capacitor has enough time to charge but not discharge

1. Figure 4.1a shows the schematic for a simple RC circuit. Simulate the circuit using PSpice as per the case II in Table 4.1.
2. Specify the “Run to Time” as 50ms to run the simulation for five full periods.
3. Attach the plot of capacitor voltage. Plots should be labelled, visible and with white background.
4. Mention the maximum charging voltage of the capacitor in Table 4.2.
5. Compare the calculated and simulated results. Is the capacitor fully charged and discharged? Justify your answer with proper reasoning.

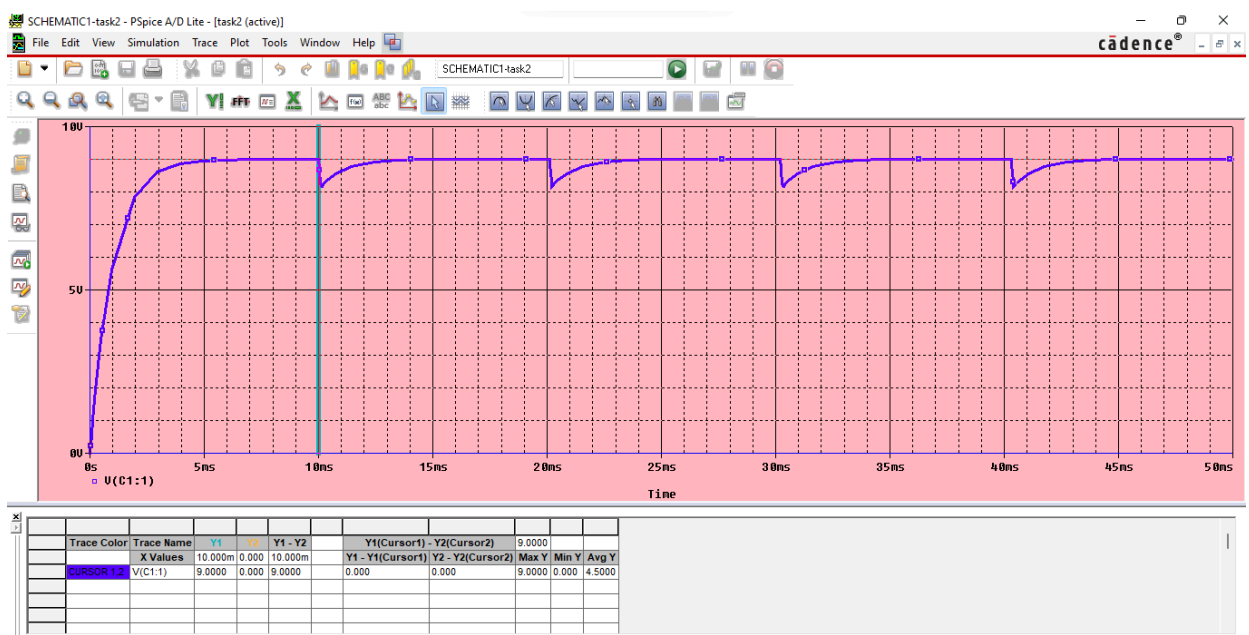
Space for Analysis:

Space for Analysis:

$$V_c(t) = V_0 e^{-t/\tau} = 9 \times e^{-\frac{0.1}{1}} = 9 \times e^{-0.1} = 8.145V \Rightarrow \text{discharging voltage.}$$

$$V_c(t) = 9(1 - e^{-10}) = 6.9996 \Rightarrow \text{charging.}$$

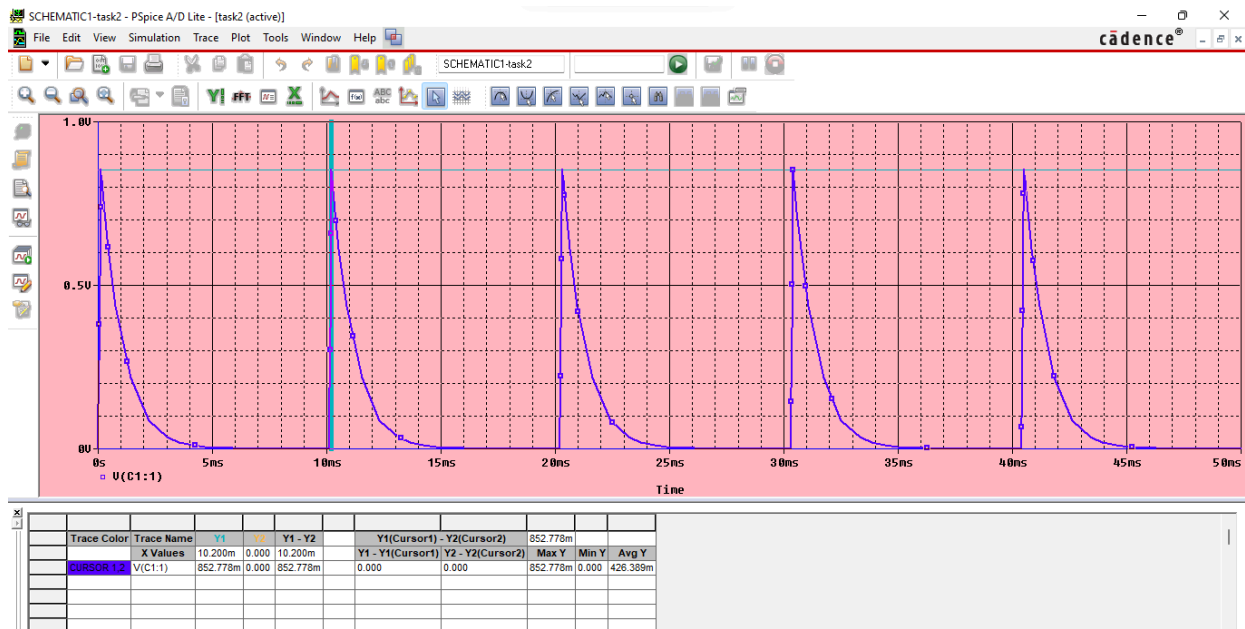
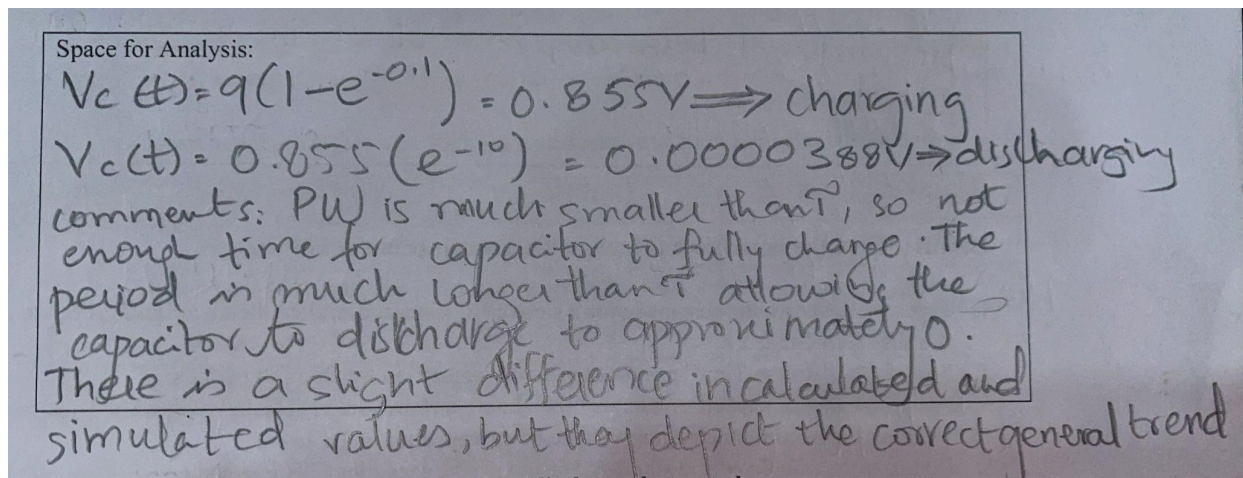
comment: since pulse width is much larger than τ , capacitor has enough time to almost fully charge to 9V. However time between pulses is very short, so capacitor doesn't have enough time to fully charge. As a result, capacitor voltage not only remains close to maximum (9V), but the minimum can also be seen to build up incrementally to maximum with each cycle.



Case III: When the capacitor has enough time to discharge but not charge

1. Figure 4.1a shows the schematic for a simple RC circuit. Simulate the circuit using PSpice as per the case III in Table 1.
2. Specify the “Run to Time” as 50ms to run the simulation for five full periods.
3. Attach the plot of capacitor voltage. Plots should be labelled, visible and with white background.
4. Mention the maximum charging voltage of the capacitor in Table 4.2.
5. Compare the calculated and simulated results. Is the capacitor fully charged and discharged? Justify your answer with proper reasoning.

Space for Analysis:

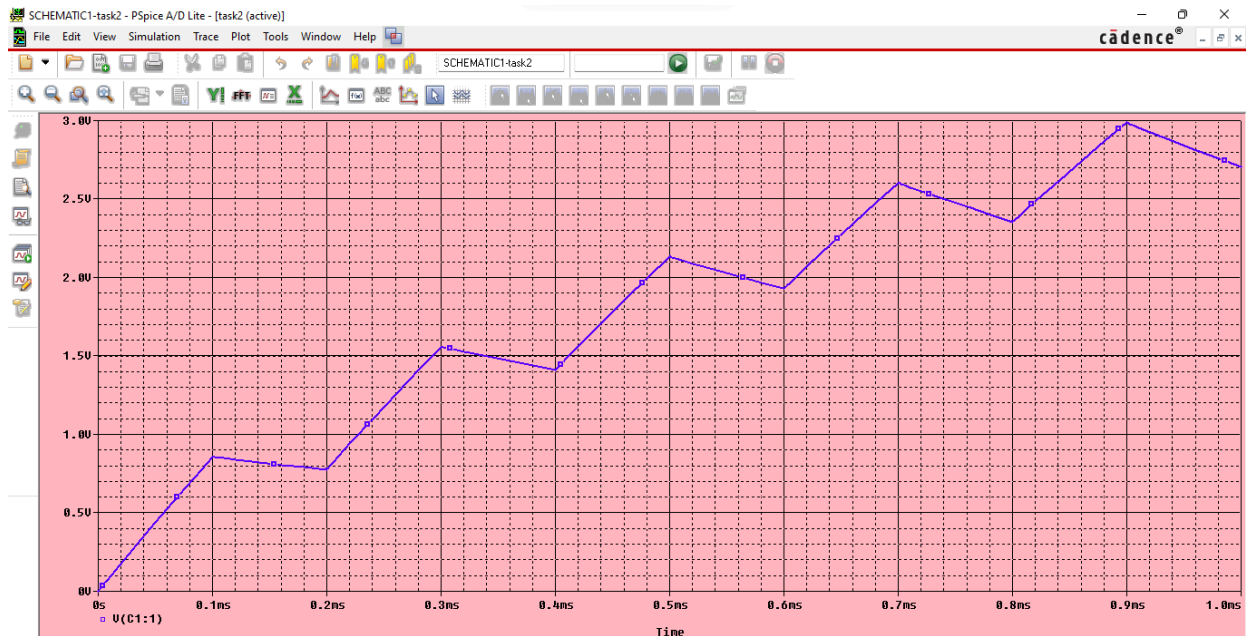


1. Figure 1a shows the schematic for a simple RC circuit. Simulate the circuit using PSpice as per the case III in Table 1.
2. Specify the “Run to Time” as 1ms to run the simulation for five full periods and then specify the “Run to Time” as 10ms run simulation for both the cases.
3. Attach the plot of capacitor voltage at both “Run to Time” values. Plots should be labelled, visible and with white background.
4. Mention the maximum charging voltage of the capacitor in Table 4.2.

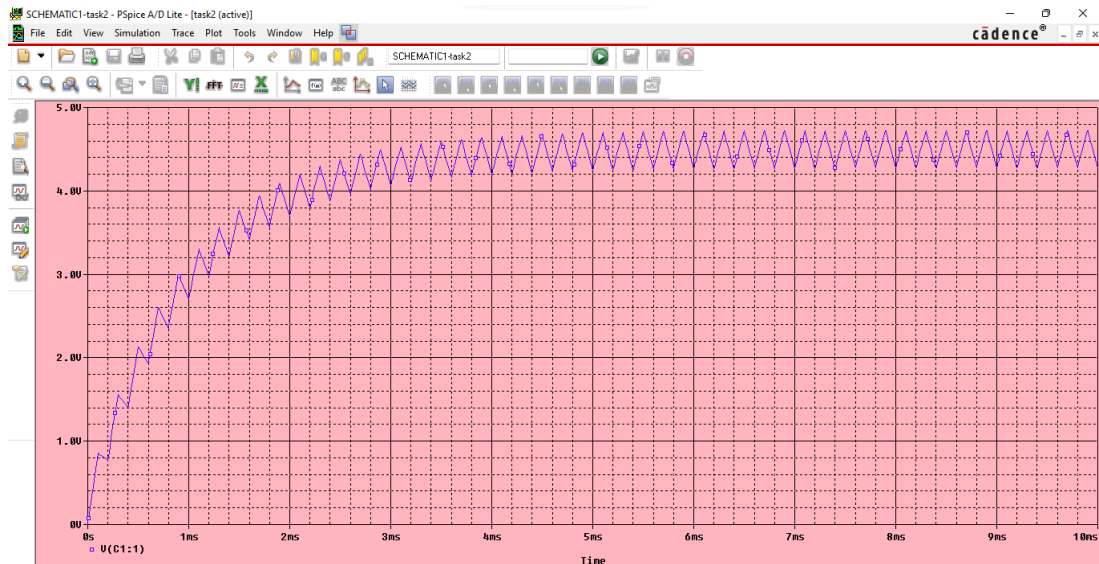


5. Compare the calculated and simulated results. Is the capacitor fully charged and discharged? Justify your answer with proper reasoning.

When run to time is 1ms:



When run to time is 10ms:



Space for Analysis:

Space for Analysis:

$$V_c = 0.855(1 - e^{-0.1}) = 0.855V \Rightarrow \text{charging}$$

$$V_c = 0.855(e^{-0.1}) = 0.774V \Rightarrow \text{discharging}$$

comments: pulse and period are much shorter than τ so the first small value peak and subsequent discharge to ~~0.774V~~ 0.774V makes sense. The erratic pattern in the simulation is because the charging and discharging cycles are so rapid.

Table 4.2: Data Table

Cases	V _c (Measured) (10 Points)	V _c (Calculated) Pre-Lab (8 Points)
Case I	8.9700	8.9996
Case II	8.9900	8.9996
Case III	0.852	0.855
Case IV	0.837	0.855

Task 3: To examine switched RC Circuits

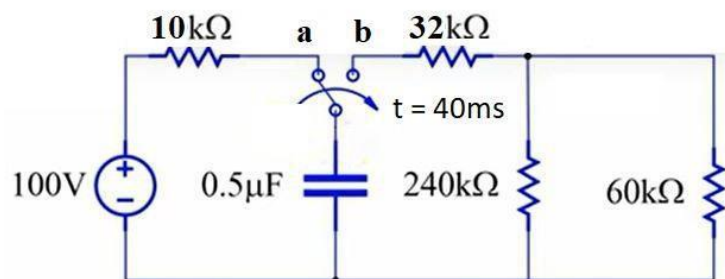
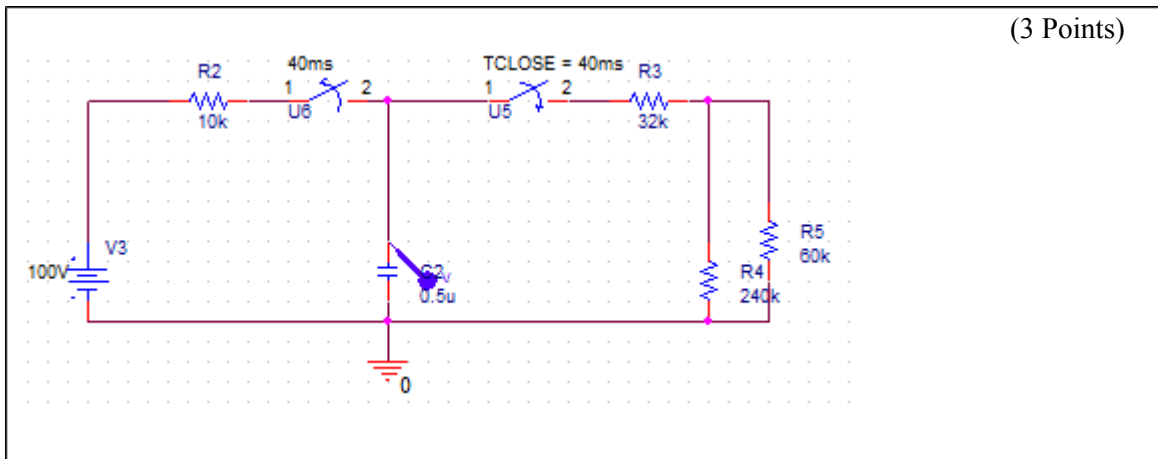


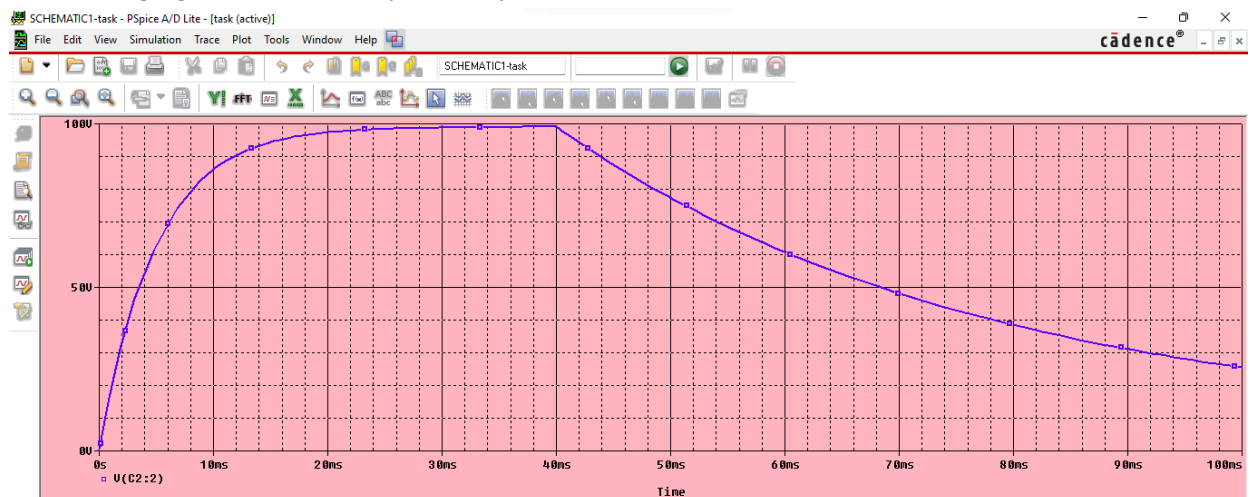
Figure 4.2

1. Simulate the circuit shown in Figure 4.2 using PSpice. Use two Single Pole Single Throw (SPST) switches to model a Single Pole Double Throw (SPDT) switch. Redraw the circuit with two SPST switches in the box below:

(3 Points)



- Set the initial condition of the capacitor to 0V, double click on the Capacitor and set IC=0. Run the simulation for 100ms.
- Attach the snippet of simulated circuit and plot of capacitor voltage. Plots should be visible and with white background.
- Discuss the results of the simulation in detail, with reference to different taus for charging and discharging, use relations in your analysis.



we can notice in the simulation that the charging and discharging time are very different. discharging takes a much longer time. this is because the value of τ changes as the circuit switches at $t = 40\text{ms}$.

$\Rightarrow (t < 40\text{ms}) \tau = (10 \times 10^3)(0.5 \times 10^{-6}) = 0.005\text{s}$

$\Rightarrow (t \geq 40\text{ms}) \tau = (80 \times 10^3)(0.5 \times 10^{-6}) = 0.04\text{s}$

we can observe that τ being greater makes discharging time more because V_c is dependant on τ using the following equation: $V_c = V_0 (e^{-t/\tau})$

Post Lab Task

Task 4: To examine switched RL Circuits

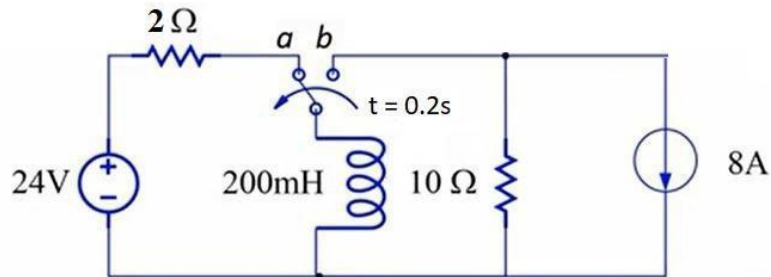
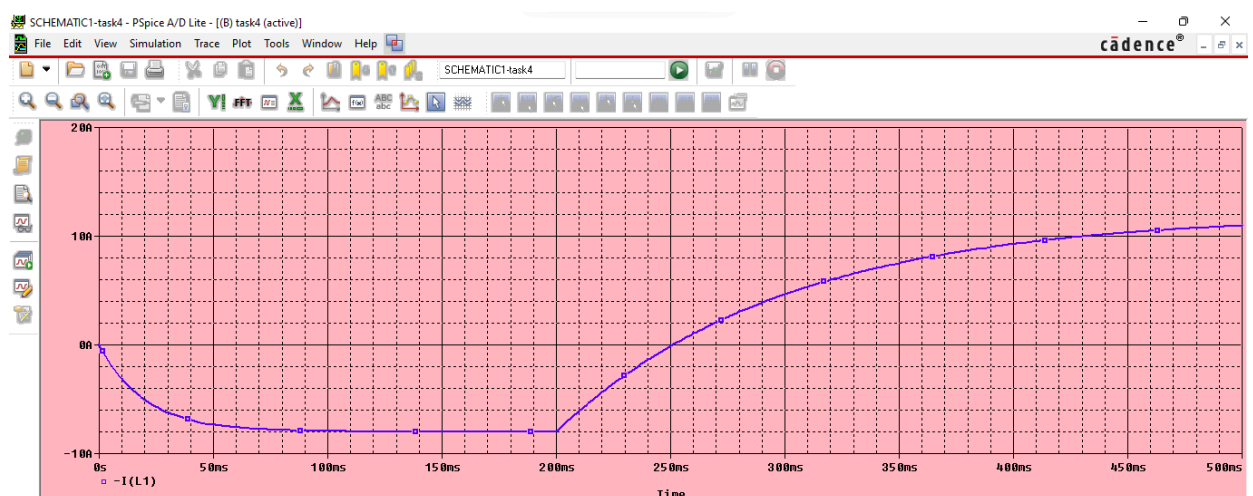
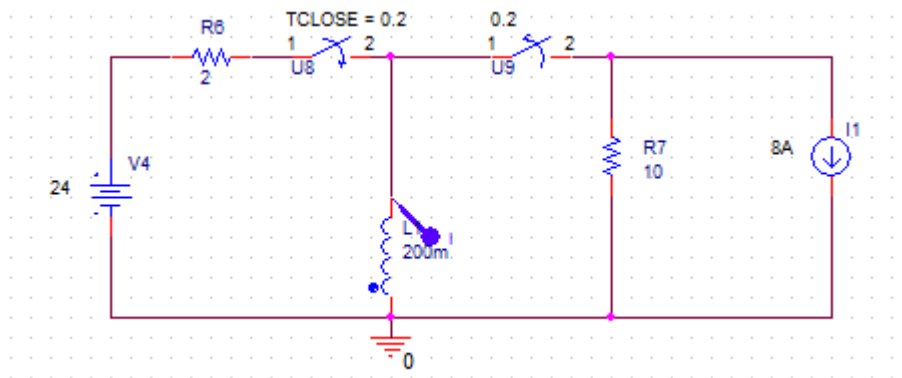


Figure 4.3

1. Simulate the circuit shown in Figure 4.3 using PSpice. Use two SPST switches to model a SPDT switch.
2. Set the initial condition of the inductor to 0A. Run the simulation for 500ms.
3. Attach the snippet of simulated circuit and plot of Inductor Current. Plots should be visible and with white background.





Frequently Asked Questions (FAQs)

Q. How is the behavior of a non-ideal inductors?

Ans: You might have always assumed that circuit elements are ideal, for example, that inductors only have inductance and not capacitance or resistance. This is generally a decent assumption, but in reality, no circuit element is truly ideal. In particular, today's "inductor" has both inductance and resistance (real inductor = ideal inductor in series with resistor). Although there is no way to physically separate the inductor from the resistor in this circuit element.



Assessment Rubric Lab 04

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Points Distribution

Task No.	LR 2 Simulation	LR5 Results/Plots	LR4 Data Collection	LR6 Calculations	LR 10 Analysis	LR9 Report
Task 1	-	-	-	/10	-	/12
Task 2	/5	/10	/10		/10	
Task 3	/5	/8			/5	
Task 4	/5	/10			-	
Total	/15	/28	/10	/10	/15	/12
CLO Mapped	CLO 1	CLO 1	CLO 1	CLO 1	CLO 1	CLO 1

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission	/10	CLO 4

CLO	Total Points	Points Obtained
1	90	
4	10	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	Circuit Layout	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	Troubleshooting	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	Calculations	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR10	Analysis	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.